

Partial Progress on Optimal One-Barrier Equilibria under Aggregation of Diverse Discount Rates

Corrected note prepared after verification of a proposed solution

June 24, 2026

Abstract

Deng, Yu and Zhang ask for a general characterization of optimal mild equilibria in a one-dimensional time-inconsistent stopping problem with aggregated discount rates. A previously circulated envelope argument does not solve this problem: saying that an equilibrium is optimal exactly when its value equals the pointwise supremum of all equilibrium values is only a restatement of the definition, unless the envelope can be computed or certified from the primitives. This note records what can be proved without that circular step. The main contribution is a complete one-barrier solution for the geometric Brownian put model with capped-linear attitude $\varphi(u) = u \wedge \alpha$ and a two-point distribution of the transformed discount exponent $Y = f(\rho)$, namely $Y = 0$ with probability p and $Y = c > 0$ with probability $1 - p$. All one-barrier equilibria are characterized, and the unique optimal one-barrier equilibrium is identified explicitly. In the low-cap regime the optimal boundary is $c/(1 + c)$, whereas the smallest equilibrium boundary is strictly smaller; hence the optimal equilibrium need not be the smallest equilibrium. The note also gives an exact semialgebraic decidability result for finite-support integer exponents and piecewise-polynomial attitude functions. These results are partial progress, not a general solution of the open problem.

1 Status of the result

The main result of this note is intentionally limited. It gives a non-circular, fully verified characterization in a concrete but nontrivial family of examples, including the capped aggregation rule used by Deng, Yu and Zhang in their examples. It does not characterize optimal equilibria for arbitrary one-dimensional diffusions, arbitrary payoff functions, arbitrary discount-rate laws, or arbitrary increasing attitude functions.

The reason for this caution is important. If $\mathcal{E}$ denotes the set of equilibria and V_R denotes the value generated by an equilibrium R , then the statement

$$R^* \text{ is optimal} \iff V_{R^*}(x) = \sup_{R \in \mathcal{E}} V_R(x) \quad \text{for all } x$$

is true, but it is only a tautology. It uses the unknown set of all equilibria to define the unknown upper envelope. The open problem is not to rename this envelope, but to characterize or certify an optimizer in terms of the model primitives. The results below avoid using the envelope as an oracle.

2 The one-barrier geometric Brownian put model

We work on the state space $[0, \infty)$. Let

$$dX_t = \mu X_t dt + \sigma X_t dW_t, \quad \mu > 0, \quad \sigma > 0,$$

and take the put payoff $g(x) = (1 - x)^+$. For $a \in [0, 1]$, write $R_a = [0, a]$ and

$$\tau_a := \inf\{t > 0 : X_t \leq a\}.$$

For each discount rate r , define $f(r) \geq 0$ by

$$\mathbb{E}_x[e^{-r\tau_a}] = \left(\frac{a}{x}\right)^{f(r)}, \quad 0 < a \leq x,$$

equivalently $f(r)$ is the nonnegative root of

$$\frac{1}{2}\sigma^2 f(f+1) - \mu f - r = 0.$$

Only the law of the transformed random variable $Y = f(\rho)$ enters the one-barrier formulas. For an increasing attitude function φ , define, for $x > a$,

$$\Lambda(x, a) := \mathbb{E} \left[\varphi \left((1 - a) \left(\frac{a}{x} \right)^Y \right) \right],$$

where the expectation is over Y . The value of the one-barrier policy R_a , provided it is an equilibrium, is

$$V_a(x) = \begin{cases} \varphi((1 - x)^+), & x \leq a, \\ \Lambda(x, a), & x > a. \end{cases}$$

The one-barrier equilibrium condition is

$$\Lambda(x, a) \geq \varphi((1 - x)^+) \quad \text{for all } x > a. \quad (1)$$

This is just the no-profitable-immediate-deviation condition for a mild equilibrium, restricted to one-barrier stopping regions.

3 A complete two-point capped-linear solution

Throughout this section assume

$$\varphi(u) = u \wedge \alpha, \quad 0 < \alpha < 1, \quad (2)$$

and

$$\mathbb{P}(Y = 0) = p, \quad \mathbb{P}(Y = c) = 1 - p, \quad 0 \leq p < 1, \quad c > 0. \quad (3)$$

Put

$$h(x) := ((1 - x)^+) \wedge \alpha, \quad s := 1 - \alpha,$$

and define the two distinguished boundaries

$$a_m := \frac{(1 - p)c}{1 + (1 - p)c}, \quad a_c := \frac{c}{1 + c}. \quad (4)$$

Here a_m is the boundary associated with the mean exponent $\mathbb{E}[Y] = (1 - p)c$, whereas a_c is the maximizer of the monomial $(1 - a)a^c$. The one-barrier continuation value is

$$\Lambda(x, a) = p((1 - a) \wedge \alpha) + (1 - p) \left((1 - a) \left(\frac{a}{x} \right)^c \wedge \alpha \right), \quad x > a. \quad (5)$$

Lemma 3.1. *Suppose $a \geq s$. Then $R_a = [0, a]$ is a one-barrier equilibrium if and only if $a \geq a_m$.*

Proof. If $a \geq s = 1 - \alpha$, then $1 - a \leq \alpha$. Hence for $a \leq x \leq 1$ both terms in (5) are uncapped and

$$\Lambda(x, a) = (1 - a) \left[p + (1 - p) \left(\frac{a}{x} \right)^c \right].$$

Since $h(x) = 1 - x$ on $[s, 1]$, and $a \geq s$, define

$$D_a(x) := \Lambda(x, a) - (1 - x), \quad a \leq x \leq 1.$$

Then $D_a(a) = 0$ and

$$D'_a(x) = 1 - (1 - p)c(1 - a)a^c x^{-c-1}.$$

The derivative $D'_a(x)$ is increasing in x , so $D_a(x) \geq 0$ on $[a, 1]$ if and only if $D'_a(a) \geq 0$. This last inequality is

$$1 - (1 - p)c \frac{1 - a}{a} \geq 0,$$

which is equivalent to $a \geq a_m$. For $x \geq 1$, $h(x) = 0 \leq \Lambda(x, a)$. Thus (1) holds exactly when $a \geq a_m$. If $a < a_m$, then $D'_a(a) < 0$, and (1) fails for all $x > a$ sufficiently close to a . $\square$

Lemma 3.2. *Suppose $a < s$. Let $M(a) := (1 - a)a^c$. Then R_a can be a one-barrier equilibrium only if $M(a) \geq M(s) = \alpha s^c$. If $s > a_c$, this condition is also sufficient.*

Proof. Take $x = s$. Since $h(s) = \alpha$ and $a < s$, the first term in (5) is capped at $p\alpha$. Therefore

$$\Lambda(s, a) = p\alpha + (1 - p) \left(\frac{M(a)}{s^c} \wedge \alpha \right).$$

If $M(a) < \alpha s^c = M(s)$, then $\Lambda(s, a) < \alpha = h(s)$, so R_a is not an equilibrium. This proves necessity.

Assume now that $s > a_c$ and $M(a) \geq M(s)$. For $a < x < s$, we have $M(a)/x^c > M(a)/s^c \geq \alpha$, so both terms in (5) are capped and $\Lambda(x, a) = \alpha = h(x)$. For $s \leq x \leq 1$,

$$\Lambda(x, a) \geq p\alpha + (1 - p)\alpha \left(\frac{s}{x} \right)^c.$$

Let

$$Q(x) := p\alpha + (1 - p)\alpha \left(\frac{s}{x} \right)^c - (1 - x), \quad s \leq x \leq 1.$$

Then $Q(s) = 0$ and

$$Q'(x) = 1 - (1 - p)c\alpha s^c x^{-c-1}.$$

This derivative is increasing in x . Since $s > a_c \geq a_m$,

$$Q'(s) = 1 - (1 - p)c \frac{\alpha}{s} \geq 0.$$

Thus $Q(x) \geq 0$ on $[s, 1]$, and hence $\Lambda(x, a) \geq h(x)$ there. Finally, $h(x) = 0 \leq \Lambda(x, a)$ for $x \geq 1$. The equilibrium condition follows. $\square$

Theorem 3.3 (All equilibria and the optimal equilibrium). *Assume (2)–(3). Let $s = 1 - \alpha$, a_m and a_c be given by (4). If $s > a_c$, let $\gamma \in (0, a_c)$ be the unique solution of*

$$(1 - \gamma)\gamma^c = \alpha s^c. \tag{6}$$

Then the one-barrier equilibria are exactly $R_a = [0, a]$ with $a \in [\underline{a}, 1]$, where

$$\underline{a} = \begin{cases} a_m, & s \leq a_m, \\ s, & a_m < s \leq a_c, \\ \gamma, & s > a_c. \end{cases} \quad (7)$$

Among these one-barrier equilibria there is a unique optimal one. Its boundary is

$$a^{\text{opt}} = \begin{cases} a_m, & s \leq a_m, \\ s, & a_m < s \leq a_c, \\ a_c, & s > a_c. \end{cases} \quad (8)$$

Thus in the low-cap regime $s > a_c$, or equivalently $\alpha < 1/(1+c)$, the optimal boundary a_c is strictly larger than the smallest equilibrium boundary γ .

Proof. We first identify the equilibrium set. Since $a_m \leq a_c$, Lemma 3.1 shows that all $a \geq \max\{s, a_m\}$ are equilibria, and that no $a \geq s$ below a_m is an equilibrium. If $a < s$, Lemma 3.2 gives the necessary condition $M(a) \geq M(s)$, where $M(a) = (1-a)a^c$. The function M is strictly increasing on $(0, a_c)$, strictly decreasing on $(a_c, 1)$, and has its unique maximum at a_c . If $s \leq a_c$, then $M(a) < M(s)$ for every $a < s$, so no equilibrium lies below s . This yields the first two cases in (7). If $s > a_c$, then the equation $M(a) = M(s) = \alpha s^c$ has exactly two roots in $[0, 1]$: $\gamma \in (0, a_c)$ and s . Lemma 3.2 shows that every $a \in [\gamma, s]$ is an equilibrium and no $a < \gamma$ is. Together with Lemma 3.1, this yields the third case in (7).

It remains to prove optimality. First suppose $s \leq a_c$, so that $\underline{a} = \max\{s, a_m\}$. Fix $x > \underline{a}$. For $a \in [\underline{a}, x]$, both terms in (5) are uncapped and

$$\Lambda(x, a) = (1-a) \left[p + (1-p) \left(\frac{a}{x} \right)^c \right].$$

Its derivative with respect to a is

$$-p + (1-p)x^{-c}a^{c-1}(c - (c+1)a).$$

Because $x \geq a$, this derivative is at most

$$-p + (1-p)a^{-1}(c - (c+1)a) = \frac{(1-p)c - a(1 + (1-p)c)}{a},$$

which is nonpositive for $a \geq a_m$. Hence $a \mapsto \Lambda(x, a)$ is nonincreasing on $[\underline{a}, x]$. If an equilibrium threshold $a \geq \underline{a}$ also satisfies $a \geq x$, then $V_a(x) = h(x) \leq V_{\underline{a}}(x)$ because $R_{\underline{a}}$ is an equilibrium. If $a < x$, the monotonicity just proved gives $V_a(x) = \Lambda(x, a) \leq \Lambda(x, \underline{a}) = V_{\underline{a}}(x)$. For $x \leq \underline{a}$, all equilibrium values equal $h(x)$ whenever $a \geq x$, and the preceding argument covers the remaining case. Therefore $R_{\underline{a}}$ is optimal in the first two regimes.

Now suppose $s > a_c$. We prove that R_{a_c} dominates every equilibrium R_a , $a \in [\gamma, 1]$. If $x < a_c$, then $V_{a_c}(x) = h(x) = \alpha$. No value can exceed α , because the attitude function itself is bounded above by α ; hence domination holds. Let $x \geq a_c$. If $a \geq x$, then $V_a(x) = h(x) \leq V_{a_c}(x)$, since R_{a_c} is an equilibrium. If $a < x$, then

$$V_a(x) = p((1-a) \wedge \alpha) + (1-p) \left(\frac{M(a)}{x^c} \wedge \alpha \right).$$

The first term is at most $p\alpha$, with equality at $a = a_c$ because $a_c < s$. The second term is at most

$$(1-p) \left(\frac{M(a_c)}{x^c} \wedge \alpha \right),$$

because $M(a) \leq M(a_c)$ for all $a \in [0, 1]$. Thus $V_a(x) \leq V_{a_c}(x)$. This proves optimality in the low-cap regime.

Uniqueness follows by taking x large. In the first two regimes the derivative estimate is strict for every $a > \underline{a}$ and large enough x , so no larger threshold can dominate $R_{\underline{a}}$. In the low-cap regime, as $x \rightarrow \infty$, the comparison of the second term is governed by the strict maximizer of $M(a)$, namely a_c . Since $1-p > 0$, no threshold $a \neq a_c$ can match R_{a_c} for all sufficiently large x . Hence the optimal one-barrier equilibrium is unique. $\square$

Corollary 3.4 (The symmetric example with a patient atom). *Let $p = 1/2$. If $\alpha < 1/(1+c)$, then the smallest one-barrier equilibrium has boundary γ , the smaller root of $(1-a)a^c = \alpha(1-\alpha)^c$, while the optimal one-barrier equilibrium has boundary $c/(1+c)$. Thus the optimal equilibrium is not the smallest equilibrium. The same conclusion holds for every $0 \leq p < 1$, not only for $p = 1/2$.*

4 A semialgebraic finite-support certificate

The previous theorem is explicit because the two-point capped-linear model has enough structure. For larger finite-support laws one should not expect a single scalar threshold such as a_m or a_c to solve the comparison problem. Nevertheless, for a large finite-dimensional class there is an exact algebraic decision procedure.

Theorem 4.1 (Semialgebraic decidability). *Assume the one-barrier geometric Brownian put model with $K = 1$. Suppose that*

$$\mathbb{P}(Y = n_i) = w_i, \quad i = 1, \dots, N,$$

where $n_i \in \mathbb{N}_0$ and the weights w_i are algebraic real numbers. Suppose also that $\varphi : [0, 1] \rightarrow \mathbb{R}$ is continuous, increasing, and piecewise polynomial with algebraic breakpoints and algebraic coefficients. Then the set

$$\mathcal{A} := \{a \in [0, 1] : R_a \text{ is a one-barrier equilibrium}\}$$

and the set

$$\mathcal{O} := \{a \in \mathcal{A} : V_a(x) \geq V_b(x) \text{ for every } b \in \mathcal{A} \text{ and every } x \geq 0\}$$

are semialgebraic subsets of $[0, 1]$. Moreover, quantifier elimination over real closed fields produces an exact finite algebraic description of $\mathcal{A}$ and $\mathcal{O}$. In particular, the existence of an optimal one-barrier equilibrium is decidable in this class.

Proof. For $x > a > 0$,

$$\Lambda(x, a) = \sum_{i=1}^N w_i \varphi \left((1-a) \frac{a^{n_i}}{x^{n_i}} \right).$$

The endpoint $a = 0$, if included, is a single separately definable case under the usual “payoff zero on $\{\tau = \infty\}$ ” convention, so it does not affect semialgebraicity. Because the exponents n_i are integers and $x > 0$, this is a piecewise rational-polynomial function of (x, a) . The graph of φ is semialgebraic, and after multiplying by positive powers of x , all inequalities involving Λ , $h(x) = \varphi((1-x)^+)$, and the value $V_a(x)$ can be written as first-order formulas in the ordered-ring language with algebraic

parameters. The cases $x = 0$, $x \leq a$, $x > a$, and $x \geq 1$ are handled by adding the corresponding polynomial inequalities and using the appropriate piece of the value definition.

Thus the assertion $a \in \mathcal{A}$ is expressed by the first-order formula

$$0 \leq a \leq 1 \quad \text{and} \quad \forall x (x > a \Rightarrow \Lambda(x, a) \geq h(x)),$$

with Λ expanded by the semialgebraic graph of φ . The assertion $a \in \mathcal{O}$ is expressed by

$$a \in \mathcal{A} \quad \text{and} \quad \forall b \forall x (b \in \mathcal{A} \Rightarrow V_a(x) \geq V_b(x)).$$

These are first-order formulas over a real closed field. By the Tarski-Seidenberg theorem, each formula is equivalent to a quantifier-free Boolean combination of polynomial equalities and inequalities with algebraic coefficients. Standard quantifier-elimination algorithms therefore give an effective exact description of $\mathcal{A}$ and $\mathcal{O}$. The emptiness of $\mathcal{O}$ is decidable from that quantifier-free description. $\square$

Remark 4.2. Theorem 4.1 should not be mistaken for a closed-form formula. Its value is that it converts the finite-support, piecewise-polynomial one-barrier problem into a finite algebraic certificate. This is materially different from defining an abstract upper envelope over all equilibria: quantifier elimination can, in principle, remove all universal quantifiers and return finitely many polynomial conditions.

5 What remains open

The results above give a complete solution only in the two-point capped-linear model and an algorithmic finite algebraic certificate in a broader finite-support class. They do not solve the open problem of Deng, Yu and Zhang in its full generality. In particular, the following issues remain outside the scope of this note:

- (i) non-barrier equilibria and the possibility that an optimal equilibrium, if it exists, is not a one-barrier region;
- (ii) arbitrary regular one-dimensional diffusions, where the hitting transform is not a monomial;
- (iii) general laws of $Y = f(\rho)$, especially continuous laws;
- (iv) general smooth strictly increasing attitude functions without the monotonicity condition on $u \mapsto u\varphi'(u)$;
- (v) structural conditions guaranteeing that the statewise maximizers of equilibrium values share a common equilibrium.

Thus this note should be read as partial progress and as a corrected replacement for a circular envelope argument, not as a full solution of the authors' open problem.

References

- [1] S. Deng, X. Yu, and J. Zhang. On time-consistent equilibrium stopping under aggregation of diverse discount rates. *Mathematics of Operations Research*, Articles in Advance, 2025. arXiv:2302.07470.

- [2] Y.-J. Huang and Z. Zhou. The optimal equilibrium for time-inconsistent stopping problems: the discrete-time case. *SIAM Journal on Control and Optimization*, 58(2):590–609, 2020.
- [3] J. Bochnak, M. Coste, and M.-F. Roy. *Real Algebraic Geometry*. Springer, 1998.